Linear Algebra with Applications in Automatic Control

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Chapter 3 Linear transformation

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For notational simplicity, $\mathcal{L}(\mathcal{V} \mapsto \mathcal{V})$ will be rewritten $\mathcal{L}(\mathcal{V})$.

From the analysis in Chapter 1, with respect to different bases of $\mathcal{V}$, the matrix representations of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$ will be different. Theorem 1.13 reveals that all the different matrix representations form an equivalent class.

- Which characteristics of linear transformation A is independent of the choice of the basis of the linear space V?
- What is the simplest form in this equivalent class?
- How is the relationship between this simplest form and the structure and the invariants of A ?

3.1 The eigenvalues of a linear transformation

Z. Song

- Recall matrix representation of a linear transformation and the associated coordinate expressions

Since $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, only one basis is needed. Denote by $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ the basis vectors. Since $\mathscr{A}(\boldsymbol{\alpha}_j) \in \mathcal{V}$, the images of the basis vectors can be again represented by the basis, and

$$\mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]) = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n] \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}$$

$$= [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n] A$$

The above equation states that the matrix representation of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$ with respect to the basis $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ is the $n \times n$ square matrix A.

3.1 The eigenvalues of a linear transformation

Z. Song

- Recall matrix representation of a linear transformation and the associated coordinate expressions

For any $\boldsymbol{\alpha} \in \mathcal{V}$,

$$\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathcal{V},$$

the image can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}) = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

The presentations of $\boldsymbol{\alpha}$ and the image $\mathscr{A}(\boldsymbol{\alpha})$ satisfy the following equation

$$\boldsymbol{y} = A\boldsymbol{x}$$

3.1 The eigenvalues of a linear transformation

Z. Song

- Applying theorem 1. 13 to the case of A belongs to $L(V)$

Theorem 3.1 Let $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\alpha_1, \alpha_2, \ldots, \alpha_n$ and $\alpha'_1, \alpha'_2, \ldots, \alpha'_n$ be two bases of $\mathcal{V}$ with transition matrix P, the representation of $\mathscr{A}$ with respect to the bases $\alpha_1, \alpha_2, \ldots, \alpha_n$ and $\alpha'_1, \alpha'_2, \ldots, \alpha'_n$ be A and A' respectively. Then $A' = P^{-1}AP$.

Definition 3.1 Let $A, B \in \mathcal{F}^{n\times n}$. If there exists $P \in \mathcal{F}^{n\times n}$ such that $B = P^{-1}AP$, then we say that B and A are similar, denoted $B \sim A$.

- The matrix representations with respect to different bases are similar.

- Similar matrices represent a unique linear transformation with respect to different bases.

3.1 The eigenvalues of a linear transformation

Z. Song

- We are interested then in the problem how to choose a proper set of basis vectors so that the representation of A has the simplest structure,

- we also care about the invariants (key properties (information) are retained)

- First introduce the concept of A-invariant subspace

Definition 3.2 Let $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\mathcal{W}$ be a subspace of $\mathcal{V}$,

$$\mathscr{A}(\mathcal{W}) \stackrel{\triangle}{=} \{\boldsymbol{\beta} \ : \ \boldsymbol{\beta} = \mathscr{A}(\boldsymbol{\alpha}), \ \boldsymbol{\alpha} \in \mathcal{W}\} .$$

If $\mathscr{A}(\mathcal{W}) \subset \mathcal{W}$, then $\mathcal{W}$ is said to be $\mathscr{A}$-invariant.

3.1 The eigenvalues of a linear transformation

Z. Song

- This section talks about invariant subspace which we call eigenspace.
- A subspace spanned the eigenvectors of a linear transformation is A-invariant., we will then discuss it in the sequel.

- First we give iff conditions of of A-invariant subspace

Theorem 3.2 Let $\mathcal{W} \subset \mathcal{V}$ be a subspace with the basis $\beta_1, \beta_2, \ldots, \beta_m$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{W}$ is $\mathscr{A}$-invariant if and only if there exists an $m \times m$ matrix $A_{\mathcal{W}}$ such that

$$\mathscr{A}([\beta_1 \ \beta_2 \ \ldots \ \beta_m]) = [\beta_1 \ \beta_2 \ \ldots \ \beta_m] A_{\mathcal{W}}$$

- Properties of A-invariant subspace, please see the first paragraph on Pg 88

3.1 The eigenvalues of a linear transformation

Z. Song

- A subspace spanned the eigenvectors of a linear transformation is A-invariant- continued

  - Definition of eigenvectors and eigenvalues

> Definition 3.3 Let $\mathscr{A} \in \mathcal{L}(\mathcal{V})$. If for $\lambda_0 \in \mathcal{F}$, there exists a nonzero vector $\boldsymbol{v}$ such that
>
> $$\mathscr{A}(\boldsymbol{v}) = \lambda_0 \boldsymbol{v} \qquad (1)$$
>
> then λ_0 is called an eigenvalue of $\mathscr{A}$, $\boldsymbol{v}$ is the eigenvector of $\mathscr{A}$ associated with the eigenvalue λ_0.

- Remark: An eigenvector is that of whose image is in the same direction with a scaling of the length.

- Implications: image of eigenvectors remains in the spanned subspace of the eigenvector, implies the eigenspace is A-invariant

3.1 The eigenvalues of a linear transformation

Z. Song

- Definition of eigenvectors and eigenvalues of a linear transformation

Definition 3.3 Let $\mathscr{A} \in \mathcal{L}(\mathcal{V})$. If for $\lambda_0 \in \mathcal{F}$, there exists a nonzero vector $\boldsymbol{v}$ such that

$$\mathscr{A}(\boldsymbol{v}) = \lambda_0 \boldsymbol{v} \tag{1}$$

then λ_0 is called an eigenvalue of $\mathscr{A}$, $\boldsymbol{v}$ is the eigenvector of $\mathscr{A}$ associated with the eigenvalue λ_0.

- Remark: the eigenvalue and eigenvector of a linear transformation A problem can be summed up to that of the corresponding matrix representation A.

3.1 The eigenvalues of a linear transformation

Z. Song

- How to find the eigenvectors and eigenvalues of a linear transformation

Let $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ be a basis of $\mathcal{V}$, the representation of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$ with respect to this basis be A.

An eigenvalue λ_0 is a root of the polynomial

$$\det(\lambda E - A).$$

- eigenvector

The representation of an eigenvector associated with the eigenvalue λ_0 is a nonzero solution to the homogeneous equation $(\lambda_0 E_n - A)\boldsymbol{x} = \boldsymbol{0}$, that is, the solution to

$$\begin{cases} (\lambda_0 - a_{11})x_1 - a_{12}x_2 - \cdots - a_{1n}x_n = 0 \\ -a_{21}x_1 + (\lambda_0 - a_{22})x_2 - \cdots - a_{2n}x_n = 0 \\ \vdots \\ -a_{n1}x_1 - a_{n2}x_2 - \cdots (\lambda_0 - a_{nn})x_n = 0 \end{cases}$$

$$\boldsymbol{v} = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{\triangleq \boldsymbol{x}}$$

3.1 The eigenvalues of a linear transformation

Z. Song

- A subspace spanned the eigenvectors of a linear transformation is A-invariant- continued
  - Definition of eigenvalue of a matrix

Definition 3.4 With respect to a matrix $A \in \mathcal{F}^{n\times n}$ and an indeterminate λ, $\lambda E - A$ is called the eigenmatrix, the determinant $\det(\lambda E - A)$ of the eigenmatrix is called the characteristic polynomial, $\det(\lambda E - A) = 0$ is called the characteristic equation, the roots of the equation are the eigenvalues (characteristic values). For an eigenvalue λ_0, the nonzero solution to equation at the bottom in the last slide is the eigenvector associated with λ_0.

- Remark: the eigenvalue and eigenvector of a linear transformation A problem can be summed up to that of the corresponding matrix representation A.

3.1 The eigenvalues of a linear transformation

Z. Song

- Based on distinct eigenvalues of a linear transformation, construct eigenspace which is A-invariant

On noting that $\mathscr{A}(\boldsymbol{v}) = \lambda_j \boldsymbol{v}$ can be equally written as $\left(\lambda_j \mathscr{E} - \mathscr{A}\right)(\boldsymbol{v}) = \boldsymbol{0}$, we can define the set

$$\mathcal{V}_{\mathrm{ev}}(\lambda_j) \triangleq \left\{ \boldsymbol{v} \ : \ \left(\lambda_j \mathscr{E} - \mathscr{A}\right)(\boldsymbol{v}) = \boldsymbol{0} \right\}$$

From the definition, if $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{V}_{\mathrm{ev}}(\lambda_j)$, then for any $k_1, k_2 \in \mathcal{F}$ there holds

$$\left(\lambda_j \mathscr{E} - \mathscr{A}\right)(k_1 \boldsymbol{v}_1 + k_2 \boldsymbol{v}_2) = k_1 \left(\lambda_j \mathscr{E} - \mathscr{A}\right)(\boldsymbol{v}_1) + k_2 \left(\lambda_j \mathscr{E} - \mathscr{A}\right)(\boldsymbol{v}_2) = \boldsymbol{0}$$

The last equation reveals that $\mathcal{V}_{\mathrm{ev}}(\lambda_j)$ is a subspace of $\mathcal{V}$.

Definition 3.5 Let $\lambda_j \ j = 1, 2, \ldots, s$ be the distinct eigenvalues of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, i.e. $\lambda_j \neq \lambda_k$ $\forall j \neq k$. $\mathcal{V}_{\mathrm{ev}}(\lambda_j)$ is referred to as the eigenspace of λ_j.

3.1 The eigenvalues of a linear transformation

- A subspace spanned the eigenvectors of a linear transformation is A-invariant- continued
- Definition of eigenspace

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$$\left(\lambda_j \mathscr{E} - \mathscr{A}\right)(k_1\boldsymbol{v}_1 + k_2\boldsymbol{v}_2) = k_1 \left(\lambda_j \mathscr{E} - \mathscr{A}\right)(\boldsymbol{v}_1) + k_2 \left(\lambda_j \mathscr{E} - \mathscr{A}\right)(\boldsymbol{v}_2) = \boldsymbol{0}$$

The last equation reveals that $\mathcal{V}_{\text{ev}}(\lambda_j)$ is a subspace of $\mathcal{V}$.

Definition 3.5 Let $\lambda_j \ j = 1, 2, \ldots, s$ be the distinct eigenvalues of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, i.e. $\lambda_j \neq \lambda_k$ $\forall j \neq k$. $\mathcal{V}_{\text{ev}}(\lambda_j)$ is referred to as the eigenspace of λ_j.

3.1 The eigenvalues of a linear transformation

Z. Song

- A subspace spanned the eigenvectors of a linear transformation is A-invariant- continued

Theorem 3.4 The eigenspace $\mathcal{V}_{ev}(\lambda_j)$ is $\mathscr{A}$-invariant. Especially, each eigenvector of $\mathscr{A}$ spans a 1-dimensional $\mathscr{A}$-invariant subspace.

- What is the dimension of the eigenspace?

Definition 3.6 Let $\mathcal{V} \simeq \mathcal{F}^n$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, λ_1, λ_2, ..., λ_s with $\lambda_j \neq \lambda_k \ \forall j \neq k$ be the distinct eigenvalues, the characteristic polynomial be

$$\det(\lambda E - A) = \prod_{j=1}^{s} (\lambda - \lambda_j)^{\hat{n}_j}$$

Then $\hat{n}_j$ is called the algebraic multiplicity of λ_j, $q_j = \dim \mathcal{V}_{ev}(\lambda_j)$ is said to be the geometric multiplicity of λ_j.

Theorem 3.3 Let $\mathcal{V} \simeq \mathcal{F}^n$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$. Then

$$\dim \mathcal{V}_{ev}(\lambda_j) = \dim \mathcal{N}(\lambda_j \mathscr{E} - \mathscr{A}) = n - \text{rank}(\lambda_j \mathscr{E} - \mathscr{A}).$$

3.1 The eigenvalues of a linear transformation

Z. Song

- Two important theorems

Theorem 3.5 Let $\lambda_1, \lambda_2, \ldots, \lambda_s$ be the distinct eigenvalues of $\mathscr{A}$ with the associated eigenvectors $\boldsymbol{v}_j$, $j = 1, 2, \ldots, s$, then $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_s$ are linearly independent.

Theorem 3.6 Let $\lambda_1, \lambda_2, \ldots, \lambda_s$ be the totality of all the distinct eigenvalues of $\mathscr{A}$, q_j be the geometric multiplicity of λ_j and $\boldsymbol{v}_{1,j}, \boldsymbol{v}_{2,j}, \ldots, \boldsymbol{v}_{q_j,j}$ form a basis for $\mathcal{V}_{\text{ev}}(\lambda_j)$. Then the eigenvectors

$$\begin{array}{llll} \boldsymbol{v}_{1,1}, & \boldsymbol{v}_{2,1}, & \ldots, & \boldsymbol{v}_{q_1,1} \\ \boldsymbol{v}_{1,2}, & \boldsymbol{v}_{2,2}, & \ldots, & \boldsymbol{v}_{q_2,2} \\ \vdots & & & \\ \boldsymbol{v}_{1,s}, & \boldsymbol{v}_{2,s}, & \ldots, & \boldsymbol{v}_{q_s,s} \end{array}$$

are linearly independent.

From theorem 3.6 we know that the sum of the s eigenspaces is a direct sum, i.e.

$$\mathcal{V}_{\text{ev}}(\lambda_1)+\mathcal{V}_{\text{ev}}(\lambda_2)+\cdots+\mathcal{V}_{\text{ev}}(\lambda_s) = \mathcal{V}_{\text{ev}}(\lambda_1)\dot{+}\mathcal{V}_{\text{ev}}(\lambda_2)\dot{+}\cdots\dot{+}\mathcal{V}_{\text{ev}}(\lambda_s)$$

3.1 The eigenvalues of a linear transformation

Z. Song

- Based on distinct eigenvalues of A matrix, we can also construct subspaces

Let A be the matrix representation of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, λ_j be an eigenvalue of $A \in \mathcal{F}^{n\times n}$,

$$\mathcal{T}_{\text{ev}}(\lambda_j) \triangleq \left\{ \boldsymbol{x} \ : \ (\lambda_j E_n - A)\boldsymbol{x} = \boldsymbol{0} \right\}$$

then $\mathcal{T}_{\text{ev}}(\lambda_j)$ is the eigenspace associated with λ_j and homogeneous to $\mathcal{V}_{\text{ev}}(\lambda_j)$. $\mathcal{T}_{\text{ev}}(\lambda_j)$ is thus a subspace of $\mathcal{F}^n$ and $\dim \mathcal{T}(\lambda_j) = q_j$, the vectors $\boldsymbol{x}_{1,j}$, $\boldsymbol{x}_{2,j}$, ..., $\boldsymbol{x}_{q_j,j}$ satisfying

$$A\boldsymbol{x}_{i,j} = \lambda_j \boldsymbol{x}_{i,j}\,, \quad i = 1, 2, \ldots, q_j$$

are the eigenvectors of the eigenvalue λ_j and form a basis of $\mathcal{T}_{\text{ev}}(\lambda_j)$.

3.1 Eigenvalues of a linear transformation

Z. Song

- The matrix representations of a linear transformation, with respect to different bases are similar.
- Properties of similar matrices

Theorem 3.7 Similar matrices have the same eigenvalues. If $\boldsymbol{x}$ is the eigenvector of A associated with the eigenvalue λ, then $P^{-1}\boldsymbol{x}$ is the eigenvector of $B = P^{-1}AP$ associated with the same eigenvalue λ. Hence, the eigenvectors of a linear transformation are independent of the choice of the basis.

3.1 The eigenvalues of a linear transformation

Z. Song

- *A belongs to L(V)*

Implications: **the eigenvalues and the eigenvectors of the linear**
•transformation are invariant with respect to different choices of the bases and can be thus described by any one of the representations.

•Based on this observation, we can study the properties of the eigenvalues and the eigenvectors of nxn matrices with scalar constant entries,
•find basis for a linear space V such that the matrix representation of a linear transformation is in its simplest form. To this end, we given the conditions on similarity.

Theorem 3.8 A, $B \in \mathcal{F}^{n\times n}$, then A is similar to B, if and only if the eigenmatrices $\lambda E - A$ and $\lambda E - B$ are equivalent.

3.1 The eigenvalues of a linear transformation

Z. Song

- A subspace spanned the eigenvectors of a linear transformation is A-invariant- continued
- An example of importance of eigenvalue and eigenvector, applications to composite transformation powers of matrices

The composite transformation $\mathscr{A}^k$ in in $\mathcal{L}(\mathcal{V})$ as well. Let the representation of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$ be A, i.e.

$$\mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]) = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]A$$

then

$$\begin{aligned} \mathscr{A}[\mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n])] &\triangleq \mathscr{A}^2([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]) \\ &= \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]A) \\ &= [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]A^2 \end{aligned}$$

that is to say the representation of $\mathscr{A}^2 \in \mathcal{L}(\mathcal{V})$ is A^2. Repeating the procedure we can readily verify that the representation for $\mathscr{A}^k \in \mathcal{L}(\mathcal{V})$ is A^k for any $k \in \mathcal{Z}_{0,+}$. Furthermore, if $\mathscr{A}(\boldsymbol{v}) = \lambda\boldsymbol{v}$, then $\mathscr{A}^k(\boldsymbol{v}) = \lambda^k\boldsymbol{v}$, which is to say that if λ is an eigenvalue of $\mathscr{A}$ with the associated eigenvector $\boldsymbol{v}$, then λ^k is an eigenvalue of $\mathscr{A}^k$ with the same associated eigenvector.

All the different matrix representations of a linear transformation form an similarity class.

- Which characteristics of A is independent of the choice of the basis of V? (from section 3.1, eigenvalues)
- What is the simplest form in this equivalent class?
- How is the relationship between this simplest form and the structure and the invariants of A ?

3.2 The Natural normal form

Z. Song

- Natural normal form via similar transformation (reduction)

Definition 2.7 The matrix N_i given below is called the column companion matrix of the polynomial $\psi_i(\lambda) = \lambda^{n_i} + \alpha_{n_i-1,i}\lambda^{n_i-1} + \cdots + \alpha_{1,i}\lambda + \alpha_{0,i} \in \mathcal{F}[\lambda]$, where

$$N_i = \begin{bmatrix} 0 & 0 & \dots & 0 & -\alpha_{0,i} \\ 1 & 0 & \ddots & \vdots & -\alpha_{1,i} \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & 1 & -\alpha_{n_i-1,i} \end{bmatrix} \in \mathcal{F}^{\nu\times\nu}$$

Theorem 3.9 $A \in \mathcal{F}^{n\times n}$ has the nonconstant invariant polynomials $\psi_1(\lambda)$, $\psi_2(\lambda)$, ..., $\psi_l(\lambda)$ with $\deg \psi_i(\lambda) = n_i$, $\psi_i(\lambda)|\psi_{i+1}(\lambda)$, $i = 1, 2, \dots, l-1$, $N_i \in \mathcal{F}^{n_i\times n_i}$ is the column companion matrix of $\psi_i(\lambda)$. Then

$$A \sim N = \text{blockdiag}\{N_1, N_2, \dots, N_l\}. \quad (1)$$

- N is the natural normal form

3.2 The Natural normal form

Z. Song

- Natural normal form via similar transformation (reduction)
  - Derivation is on Pg 96, the key points:

Let $n_i = \deg \psi_i(\lambda),\ i = 1, 2, \ldots, l,$

$$\lambda E_n - A \cong \text{diag}\left\{ E_{n_1-1}, E_{n_2-1}, \ldots, E_{n_l-1}, \psi_1(\lambda), \psi_2(\lambda), \ldots, \psi_l(\lambda) \right\}.$$

$$\text{diag}\left\{ E_{n_1-1}, \psi_1(\lambda), E_{n_2-1}, \psi_2(\lambda), \ldots, E_{n_l-1}, \psi_l(\lambda) \right\}$$

Denote $S_i(\lambda) = \text{diag}\{ E_{n_i-1}, \psi_i(\lambda) \}$, then

$$\lambda E_n - A \cong \text{blockdiag}\{ S_1(\lambda), S_2(\lambda), \ldots, S_l(\lambda) \}.$$

It is clear that $S_i(\lambda) \in \mathcal{F}^{n_i \times n_i}[\lambda]$ is a diagonal λ-matrix and its only nonconstant diagonal element $\psi_i(\lambda)$ is of degree n_i and at the bottom. Hence, $S_i(\lambda)$ is in Smith canonical form with the only nonconstant invariant polynomial $\psi_i(\lambda)$.

3.2 The Natural normal form

Z. Song

- Natural normal form via similar transformation (reduction)
  - Derivation is on Pg 96, the key points:

Recalling example 2.26 and the symmetry of the equivalence of λ-matrices, we obtain

$$S_i(\lambda) \cong \lambda E_{n_i} - N_i$$

where

$$N_i = \begin{bmatrix} 0 & 0 & \dots & 0 & -\alpha_{0,i} \\ 1 & 0 & \ddots & \vdots & -\alpha_{1,i} \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & 1 & -\alpha_{n_i-1,i} \end{bmatrix} \in \mathcal{F}^{\nu\times\nu} \quad (1)$$

with

$$\psi_i(\lambda) = \lambda^{n_i} + \alpha_{n_i-1,i}\lambda^{n_i-1} + \cdots + \alpha_{1,i}\lambda + \alpha_{0,i}\,,$$

3.2 The Natural normal form

- Natural normal form via similar transformation (reduction)
  - Derivation is on Pg 96, the key points:

Hence

$$\begin{aligned} & \text{blockdiag}\{ S_1(\lambda), S_2(\lambda), \ldots, S_l(\lambda) \} \\ \cong\ & \text{blockdiag}\{ \lambda E_{n_1} - N_1, \lambda E_{n_2} - N_2, \ldots, \lambda E_{n_l} - N_l \} \end{aligned}$$

Denote

$$N = \text{diag}\{ N_1, N_2, \ldots, N_l \}$$

then from the transitivity of the equivalence of λ-matrices follows

$$\lambda E_n - A \cong \text{blockdiag}\{ S_1(\lambda), S_2(\lambda), \ldots, S_l(\lambda) \} \cong \lambda E_n - N$$

From theorem 3.8 we have $A \sim N$.
Since N is block diagonal and the block diagonals have the simple structure the form of N_i, we have thus found a simple form for A in its similarity class.